

**LUBRICANTS FOR
ENVIRONMENTALLY
SENSITIVE AREAS**

LUBRICANTS FOR ENVIRONMENTALLY SENSITIVE AREAS

Combining economy and ecology in a meaningful way is a topic that has been of concern to people in our societies for decades and is now of great and now also very high importance. Working in sensitive ecosystems can possibly lead to considerable damage to the environment and high follow-up costs if the wrong lubricants are used. In order to be able to support you as a customer in this challenge, lubricants have been developed for the environmentally sensitive area. These combine the idea of ecology and economy, are easily biodegradable according to OECD degradability test and are technically mature. The lubricants developed by Zeller+Gmelin for the environmentally sensitive area are in no way inferior to mineral oil-based products in terms of performance and economy and find an equally broad field of application on the market.

Experience makes it possible: Zeller+Gmelin offers you a wide range of lubricants for the environmentally sensitive area –we have put together a portfolio for you with saw chain oils, special hydraulic oils, greases, building release agents and other products.

We are certified according to DIN EN ISO 9001:2015, DIN EN ISO 14001:2015 and DIN EN ISO 50001:2018 and guarantee you a constantly high quality standard of our products.

The term “biolubricant” or “bio” refers to a corresponding environmental label or a corresponding OECD 301 test.

SAW-CHAIN OIL + EQUIPMENT CLEANER

Divinol Bio*-Kettenöl R

21820

Easily biodegradable high-performance saw-chain oil based on renewable raw materials. It can be used all year round. Excellent adhesion, wear and corrosion protection.

Specifications/Recommendations/Awards:
KWF-Test; Blue Angel according to RAL-UZ 178; EU Ecolabel;
The Austrian Ecolabel



Divinol Bio*-Chain Oil RF

26160

Biodegradable high-performance saw chain oil with resin protection. High wear protection, optimal adhesive properties and economical consumption. DIN-tested BIOBASED by DIN CERTCO.

Specifications/Recommendations/Awards:
OECD 301 B

Divinol Bio*-Kettenöl HV

31780

Biodegradable high-performance saw-chain oil based on renewable raw materials. Specially developed for high-performance chainsaws of wood harvesters. High viscosity and lubricating properties.

Specifications/Recommendations/Awards:
KWF-Test; Blue Angel according to RAL-UZ 178



Divinol Gerätereiniger Forte

12300

It quickly and reliably removes heavy oily, greasy and resinous dirt. Universally applicable. Do not apply on glass or transparent surfaces!

Specifications/Recommendations/Awards:
OECD 301 B

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HYDRAULIC OILS

Divinol HE ISO 32

32150

Synthetic, biodegradable ester product based on a powerful, environmentally friendly additive combination. The product has an excellent viscosity-temperature behaviour and can therefore be used all year round. Applications are e.g. hydraulic systems which are mainly operated outdoors, which are exposed to large temperature fluctuations and where environmental pollution due to oil leaks must be avoided, such as in mobile cranes and excavators, in agriculture and forestry, floating dredgers, lock hydraulics, ski slope equipment, conveyor systems, etc. Divinol HE ISO 32 has excellent properties in terms of oxidation stability, corrosion resistance, low temperature and EP behaviour.

Specifications/Recommendations/Awards:
DIN ISO 15380 HEES; DIN 51524-3; HVLP; EU Ecolabel; VDMA 24568

Viscosity at 40 °C mm²/s: 32 – suitable for the viscosity range from ISO 22 to 46



Divinol HE ISO 46

48800

Synthetic, easily biodegradable, partially saturated ester product based on a high-performance, environmentally friendly additive combination. The product has an excellent viscosity-temperature behaviour and can therefore be used all year round. Applications are, e.g. hydraulic systems that are mainly operated outdoors, which are exposed to large temperature fluctuations and environmental pollution due to oil leaks must be avoided. Possible applications are, e.g. in mobile cranes and excavators, in agriculture and forestry, floating dredgers, lock hydraulics, ski slope equipment, conveyor systems etc. Divinol HE ISO 46 has excellent properties with regard to oxidation stability, corrosion resistance, low temperature and EP behaviour.

Specifications/Recommendations/Awards:
DIN ISO 15380 HEES; DIN 51524-3; HVLP; VDMA 24568; OECD 301 B

Viscosity at 40 °C mm²/s: 46 – suitable for the viscosity range from ISO 32 to 68

Divinol Bio*-Hydraulik ISO 46

28370

Synthetic, easily biodegradable, saturated ester product (long oil change intervals) based on a high-performance, environmentally friendly additive combination. The product has an excellent viscosity-temperature behaviour and can therefore be used all year round. Applications are, e.g. hydraulic systems that are mainly operated outdoors, which are exposed to large temperature fluctuations and environmental pollution due to oil leaks must be avoided. Possible applications e.g. in mobile cranes and excavators, in agriculture and forestry, floating dredgers, lock hydraulics, ski slope equipment, conveyor systems etc. Divinol Bio*-Hydraulik ISO 46 has excellent properties with regard to oxidation stability, corrosion resistance, low temperature and EP behaviour.

Specifications/Recommendations/Awards:
Exceeds VDMA 24568, VDMA 24570; Swedish Standard SS 15 54 34 category V (Dry ASTM D943); Eaton Brochure 03-401-2010 (Dry ASTM D943); JCMAS P042 HKB; Parker Denison HF-1, HF-2, HF-6 (Dry ASTM D4310); USDA Biopreferred®, US Vessel General Permit (VGP); EU Ecolabel; The Austrian Ecolabel; OECD 301 B

Viscosity at 40 °C mm²/s: 46 – suitable for the viscosity range from ISO 32 to 68



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GREASES

Divinol Ecogrease LC 1 EP30470

Biodegradable grease for rolling and sliding bearing applications in a wide temperature range. Divinol Ecogrease LC 1 EP is further characterised by its advantageous delivery behaviour even at low temperatures. It is particularly suitable for use in agricultural and forestry applications as well as for the construction industry and water management.

- Specifications/Recommendations/Awards:**
- + synthetic lithium lime soap grease based on ester oil
 - + very good wear and corrosion protection and pronounced walk stability
 - + water and oxidation resistant
 - + NLGI class 1
 - + marking according to DIN 51825: KPF 1 N-40
 - + biodegradable according to DIN EN 16807 (OECD 301 B >50%)

Divinol Ecogrease LC 2 EP28130

Easily biodegradable grease for rolling and sliding bearing applications in a wide temperature range. Divinol Ecogrease LC 2 EP is further characterised by its advantageous delivery behaviour even at low temperatures. It is particularly suitable for use in agricultural and forestry applications as well as for the construction industry and water management.

- Specifications/Recommendations/Awards:**
- + synthetic lithium lime soap grease based on ester oil
 - + very good wear and corrosion protection and pronounced walk stability
 - + water and oxidation resistant
 - + NLGI class 2
 - + marking according to DIN 51825: KPF 2 N-30
 - + OECD 301 B

Divinol Fett R 295850

Easily biodegradable, water-repellent grease based on rapeseed oil for low-thermal lubrication points with loss lubrication and with regular re-lubrication. It is used, for example, in water management, agriculture and forestry.

- Specifications/Recommendations/Awards:**
- + calcium saponified grease based on rapeseed oil
 - + corrosion and water resistant
 - + good anti-wear properties
 - + NLGI class 2
 - + marking according to DIN 51825: K 2 E-20
 - + OECD 301 F

RAILWAY LUBRICANTS

Divinol Trackswitch Grease W27930

Easily biodegradable product, is used for the lubrication of switches. The sliding chairs and plates of the lubricated switches are reliably protected against wear and corrosion. The excellent low-temperature behaviour ensures sufficient lubrication capacity at low temperatures.



- Specifications/Recommendations/Awards:**
- + calcium saponified grease on synthetic ester basis
 - + with high UV-resistance
 - + corrosion and water resistant
 - + it has a good cold behaviour and good pumpability
 - + OECD 301 B



CERTIFICATE

ZELLER + GMELIN GMBH & CO. KG

has balanced the **CO₂-EMISSIONS** of the companies Zeller + Gmelin GmbH & Co. KG, Südöl Mineralöl-Raffinerie GmbH, Südöl Umwelt Service – Recycling GmbH und ZG Fluidmanagement GmbH for **2020** through the purchase of **6.000 CERTIFICATES** from the projects “GS Biomass India+ Wind Turkey + UN CER Wind China ”.

17.12.2019

DATE

FOKUS
ZUKUNFT



SIGNATURE

FOKUS-ZUKUNFT GMBH & CO. KG

www.fokus-zukunft.com

ENTERPRISE + ENVIRONMENT

The way we see it, they're inextricably linked.

Because as an independent and recession-proof medium-sized company, Zeller+Gmelin represents healthy and sustainable growth. For us, this means doing business in a way that is socially and environmentally responsible. Maintaining and improving our performance always takes priority over maximising profit in the short term. That's why we don't just care about our economic success – we also look after our employees and the environment.

The careful use of resources and the responsibility we have to future generations are integral to our philosophy. We therefore work with a certified environmental management system and an environmentally friendly supply chain. This enables us to protect our employees and facilities effectively from harmful conditions. In addition, our research and development department is continuously working on innovative solutions to ease the burden on the environment and improve processes for our customers.

Our subsidiary SÜDÖL plays a vital role in this. With an ultramodern fleet of vehicles for collecting waste oils, machining emulsions and workshop waste, we take care of the complete recovery programme for workshops, filling stations and industry – cleanly, reliably and on time. We have our own dedicated treatment plant for resource-saving and environmentally friendly recycling – made possible thanks to high-quality products and innovative processes. This allows us to offer our customers safe disposal and to close the resource cycle.

With our example and our guidelines, we encourage all our employees to make the issues of environment, health and safety an integral part of their day-to-day work. We also promote economic and social well-being wherever we do business. With regular environmental awards for resource efficiency and sustainability, we know we are on the right track.



LUBRICANTS FOR ENVIRONMENTALLY SENSITIVE AREAS



RELEASE AGENTS FOR CONCRETE AND ASPHALT

Divinol Syn BE Plus

31860

Divinol SYN BE Plus is used for all types of formwork commonly used in building construction and civil engineering: For non-absorbent or low-absorbent formwork (phenolic resin shuttering skins etc.) and for all-plastic and plastic-coated formwork skins. The use on heated formwork is possible. The product is ideal as a mixer protection and is biodegradable.

Specifications/Recommendations/Awards:

- + Blue Angel according to RAL U-UZ 178
- + EU Ecolabel
- + BTM 10 free of labeling

Viscosity at 20 °C mm²/s: 15



Divinol Asphalt MP

27430

Divinol Asphalt MP was specially developed for stone mastic asphalt (polymer modified asphalt – PMB) as well as for open porous asphalt (OPA – whisper asphalt). The product is also suitable for conventional binder courses and asphalt, and for asphalt mixing plants. Divinol Asphalt MP leaves no stains/residues on concrete and asphalt or does not dissolve asphalt. There is also a load security for asphalt troughs/tippers, as the loading areas remain free of oily residues.

Specifications/Recommendations/Awards:

- + water-miscible asphalt release agent
- + on a mineral oil-free basis
- + odourless
- + OECD 301 C

Viscosity at 20 °C mm²/s: 8

Divinol High Liquid Plus

20890

Divinol High Liquid Plus can be used to produce high-quality concrete parts by using special base oils. The product can be used for all types of formwork in building construction and civil engineering. Particularly suitable for GRP formwork skin as well as for polyurethane formwork (Reckli, NOE etc.) Divinol High Liquid Plus can also be used as mixer protection in devices, construction and transport vehicles. The use of Divinol High Liquid Plus guarantees easy and clean formwork removal as well as optimal surface results.

Specifications/Recommendations/Awards:

- + non-water-miscible, solvent-free concrete release agent
- + for high quality exposed concrete parts up to SB 3
- + reduces pore and void formation
- + OECD 301 B

Viscosity at 20 °C mm²/s: 7

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ZELLER+GMELIN

EXPERTLY DONE.

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